

Normalization Report: P2 Bank Management System

Overview

This document outlines the database normalization process for the Bank Management System, ensuring the database is designed up to the Third Normal Form (3NF) / Boyce-Codd Normal Form (BCNF). Normalization minimizes data redundancy and eliminates insertion, update, and deletion anomalies.

1. Unnormalized Form (UNF)

In the Unnormalized Form, data may contain repeating groups or multi-valued attributes. A simple view of customer transactions could look like this:

Table: BankData_UNF

- CustomerID , CustomerName , Email , Phone , Address
- AccountID , AccountType , Balance
- TransactionID , TransactionType , Amount , TransactionDate

Anomalies in UNF:

- Update Anomaly:** Updating a customer's address requires changing multiple rows if they have multiple accounts/transactions.
 - Insertion Anomaly:** Cannot add a customer without an account.
 - Deletion Anomaly:** Deleting a transaction might delete the only record of an account or customer.
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2. First Normal Form (1NF)

Rule: Eliminate repeating groups. Every column must hold atomic (indivisible) values, and each record must have a unique identifier (Primary Key).

Resolution:

We separate the compound attributes (like Address into City, State, Zip) and ensure no arrays or comma-separated values exist.

Tables in 1NF:

- **Customers:** CustomerID (PK), FirstName , LastName , Email , Phone , AddressLine , City , State , ZipCode
 - **Accounts:** AccountID (PK), CustomerID , AccountType , Balance , Status , OpenedDate
 - **Transactions:** TransactionID (PK), AccountID , TransactionType , Amount , TransactionDate , ReferenceAccountID
 - **AuditLogs:** LogID (PK), ActionType , TableName , RecordID , OldBalance , NewBalance , ActionDate
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3. Second Normal Form (2NF)

Rule: Must be in 1NF and have no partial dependencies. All non-key attributes must be fully functionally dependent on the entire Primary Key.

Resolution:

Since all of our proposed tables use single-column surrogate primary keys (CustomerID , AccountID , TransactionID , LogID), partial dependencies (which only occur with composite primary keys) are inherently impossible.

Therefore, our 1NF tables are automatically in 2NF.

4. Third Normal Form (3NF) & BCNF

Rule: Must be in 2NF and have no transitive dependencies. Non-key attributes must not depend on other non-key attributes. (BCNF further requires that every determinant must be a candidate key).

Resolution:

Let's examine our tables for transitive dependencies:

- **Customers:** FirstName , LastName , Email , Phone , etc., only depend directly on CustomerID . (City/State could technically depend on ZipCode, but in standard practice for basic addresses, this is considered acceptable or resolved via a separate ZipCode lookup table. For this project's scope, the current structure is standard 3NF).
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- **Accounts:** AccountType , Balance , Status , OpenedDate depend strictly on AccountID .
- **Transactions:** TransactionType , Amount , TransactionDate , ReferenceAccountID depend strictly on TransactionID .
- **AuditLogs:** All audit details depend purely on LogID .

Final 3NF/BCNF Schema:

1. Customers

- CustomerID (Primary Key)
- FirstName , LastName , Email , Phone , AddressLine , City , State , ZipCode

2. Accounts

- AccountID (Primary Key)
- CustomerID (Foreign Key referencing Customers)
- AccountType , Balance , Status , OpenedDate

3. Transactions

- TransactionID (Primary Key)
- AccountID (Foreign Key referencing Accounts)
- TransactionType , Amount , TransactionDate , ReferenceAccountID

4. AuditLogs

- LogID (Primary Key)
- ActionType , TableName , RecordID , OldBalance , NewBalance , ActionDate

Conclusion

The resulting database design enforces strict 1-to-Many relationships:

- One Customer can have Multiple Accounts (1-M).
- One Account can have Multiple Transactions (1-M).

This schema successfully satisfies 3NF/BCNF requirements, guaranteeing optimal data integrity and efficient querying.